AVS Hull

User's Manual

This is a preliminary user's manual. I will continue to add details as I have time.

1 General note on how the program works

AVS Hull is a web-based program. But even though it is "web based", it runs entirely inside your web browser on your machine. No part of the program runs on the AVS server. What this means:

1. When you save designs or layouts, they are stored on your computer, not "in the cloud".
2. Web browsers are somewhat restrictive on what they allow you to do with files. Depending on how you have your browser configured, your files may be saved into a "downloads" directory, or it may ask you where you want to save files. This is a feature of your browser, not an AVS Hull feature.

2 Basic Principles

The shape of the hull is defined by bulkheads. You control the number and shapes of the bulkheads. Everything else about the hull is determined by the bulkheads. Bulkheads come in three forms:

1. Bow: Can come at either the front or back of the boat (bow or stern). This bulkhead has all of its points on the centerline. It is used for "pointy" bows (and sterns).
2. Vertical: All bulkheads except the bow and stern are vertical. They are always straight up and down (thus "vertical"), and they are symmetric left and right.
3. Stern: Can come at either the front or back of the boat (bow or stern). This bulkhead type is similar to a vertical bulkhead except that you get to specify an angle. It does not have to be vertical.

The number of bulkheads, and the number of points on each bulkhead (which become "chines") is specified when you initially create the hull (but these values can be changed later).

There are three main screens in the program: Design, Layout, and Waterlines. The design screen is used to create the shape and size of your hull. The Layout screen is used to position the panels onto sheets of plywood. This will tell you what shapes to cut to create your hull. The Waterlines screen is used to compute and display various attributes of the hull (such as how much weight can it hold before it sinks). These three screens are described in the following sections.

Note on units:

AVS Hull does not assume anything about the units of numbers. I tend to think in inches (because I'm an American), but if I wanted to create a hull for my metric friends, I don't need to check any box to say the units are now meters (or centimeters (or kilometers, for that matter)). I just need to enter values that are useful to metric people. For example, instead of a 120 inch long boat (10 feet), I could enter 9 for the length, call it meters, and I now have a boat of approximately the same length but in different units.

The one place where the units matter is on the Waterlines tab. See the Waterlines section of this document for an explanation of the units on that screen and how to correct for different hull units.

A note on coordinates:

The X coordinate measures the distance from the centerline of the hull outward. It is the unit of width.

The Y coordinate measures the distance from the lowest point on the hull up. It is the unit of height.

The Z coordinate measures the distance from the bow of the hull back. It is the unit of length.

3 Program use

This section describes the three screens that are used to design your hull.

3.1 Design

The Design screen is used to create the shape and size of your hull. When you first launch AVS Hull, it will display a default hull. The parameters for this hull are based on the parameters you used the last time you created a hull.

3.1.1 Editing the hull shape

The design screen shows four views of the hull: front, side, top, and variable. The front, side, and top views are fixed – you can view them, but you can't change them. The variable view allows you to edit the shape of the hull and to rotate the hull to see it from different angles.

To rotate the hull, click inside the variable window and drag the mouse (with the button clicked). The location of your click and the direction of your drag control how the hull rotates. In general, if you click near the center (left-right) and drag up-down, the hull rotates on an axis the runs left-right through the hull. If you click on the sides and drag left-right, the hull rotates on an axis that extends vertically out of the screen. You can change the drag angle to affect the axis of rotation.

The variable window has two modes: view only and edit. The color of the hull indicates which mode you are in. To put this window into edit mode, double click on one of the other three windows. The view you clicked on will appear in the variable window, and the color of the hull will indicate that you are in edit mode. The hull will remain in edit mode until you rotate it.

3.1.2 File menu

The file menu has four options: Open, Save, Export to XML, and Create

3.1.2.1 Open

The Open menu item brings you to a screen where you can select what file to open. The file should be a file that you previously saved with the Save menu. You could also open a file provided by someone else, or a file edited by hand provided the file had the proper format (see the section on file formats later in this document).

When you open a file, any data in the Layout tab will be invalidated. It won't be deleted, but the next time you go to the Layout tab, you will be asked if you want to reload the panals.

3.1.2.2 Save

The save menu item allows you to save your designs so you can edit them later. The behavior of this item will vary depending on the configuration of your browser. Some browsers are configured so that all downloaded files go to a specific "downloads" directory. If you configure your browser so that it asks you where to save files (recommended), then when you select "Save", you will be shown a dialog box where you get to save the file. The default name of the file will be the name of the hull you specified when creating the hull (or "unnamed" if you did not specify a name). The extension on the file should not be changed: It should be ".avsh" (as in AVS Hull).

3.1.2.3 Export to XML

This is another way to save the details about your hull. It might be useful for importing the data to other programs.

Note: This menu item may be removed in future versions of AVS Hull.

3.1.2.4 Create

This is where you get to set the starting parameters for your hull. You can set the following parameters:

1. Name: This name is used as the default name for file when you Save (either designs or layouts).
2. Bow bulkhead type. You have the following options:
    a. Bow: All points are on the centerline of the hull. This is used to create a "pointy" end on your boat.
    b. Vertical: The bulkhead is straight up and down. The points are not on the centerline.
    c. Transom: The bulkhead is like a vertical bulkhead, but you can specify an angle. An angle of 90 is vertical.
3. Bow transom angle: This field is only used if the Transom bulkhead type is selected for the bow.
4. Stern bulkhead type: See description for Bow bulkhead type.
5. Stern transom angle: See description for Bow transom angle.
6. Overall length: Total length of the hull.
7. Overall width: Approximate width of the hull. The value is approximate because the curves that define the chines may extend beyond to width of the widest bulkhead. If you want a precise width, create the hull then use the Resize feature in the Edit menu to resize your hull to the correct width.
8. Overall height: Overall height of the hull.
9. Number of bulkheads (including the bow and stern)
10. Number of chines. This indicates how many panels will be used to form each side of the hull (but see Flat Bottomed, below).
11. Closed top: This is for hulls such as kayaks. This feature is not implemented in the current version of AVS Hull, but should be in a future version.
12. Flat Bottomed. If this box is checked, the hull will have a flat bottom (obviously). This will cause an extra panel to be generated, which is the bottom. For example, a flat bottomed hull with two chines will have two panels for each side, plus the bottom. If this box is not checked, there will be a seam at the center of the hull. A hull formed with two chines that is not flat bottomed would only have two panels per side. The two inner side panels will meet at the centerline of the hull.

3.1.3 Edit menu

To be described later.

3.1.3.1 Resize

To be described later.

3.1.3.2 Chines

To be described later.

3.1.3.3 Add Bulkhead

To be described later.

3.1.3.4 Delete Bulkhead

To be described later.

3.2 Layout

The Layout screen is used to create a layout of what panels need to be cut in order to create your hull. You can control the size, number, and layout of the sheets (of plywood) that you will use as well as what panels go where.

More details on how to use this screen will be included in future versions of this user manual.

3.3 Waterlines

The Waterlines screen is used to compute various parameters about your hull.

More details on how to use this screen will be included in future versions of this user manual.

4 Technical Details

4.1 File Formats

To be described in a future version of this User's Manual.